

DeBug : A Decentralised Bug-Bounty Platform

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Abstract— Bug bounty programs have emerged as an effective method for identifying software vulnerabilities through ethical hacker participation. However, existing bug bounty platforms rely heavily on centralized infrastructures, leading to challenges such as delayed payouts, opaque validation processes, privacy concerns, and limited transparency in vulnerability handling. This paper proposes DeBug, a decentralized bug bounty platform that leverages blockchain technology, smart contracts, and decentralized storage to establish a transparent and trustless vulnerability disclosure ecosystem. The platform utilizes Solana-based smart contracts to automate bounty escrow management and payout distribution. Vulnerability reports are securely stored using InterPlanetary File System (IPFS), ensuring tamper-proof storage and decentralized accessibility. Additionally, the system integrates Zero-Knowledge identity verification to allow pseudonymous participation while maintaining authenticity. A decentralized reputation system further incentivizes ethical hackers by enabling cross-program credibility tracking. The proposed architecture combines blockchain transparency with scalable off-chain infrastructure to create a secure and efficient vulnerability management ecosystem.

Keywords— Blockchain Security, Bug Bounty Programs, Smart Contracts, IPFS, Decentralized Applications, Cybersecurity

I. INTRODUCTION

The rapid growth of digital infrastructures has significantly increased the importance of cybersecurity in modern organizations. Software vulnerabilities remain one of the most critical attack vectors exploited by malicious actors. Bug bounty programs have emerged as a widely adopted approach to identifying vulnerabilities by incentivizing independent security researchers to report security flaws responsibly. Platforms such as HackerOne and Bugcrowd have demonstrated the effectiveness of crowdsourced vulnerability discovery. However, several studies highlight fundamental limitations in centralized bug bounty platforms, including lack of transparency in reward distribution, privacy risks for researchers, and inefficient dispute resolution mechanisms.

Research conducted by Hoffman et al. proposes Bountychain, a blockchain-based bug bounty system designed to automate reward payments and ensure transparency through smart contracts [1]. Similarly, the VeriOSS framework introduces blockchain-enabled vulnerability disclosure mechanisms to guarantee fair incentive distribution [2]. These approaches demonstrate the potential of blockchain technology in improving the bug bounty ecosystem.

Despite these advancements, existing blockchain-based bug bounty platforms face challenges related to scalability, high transaction costs, and limited privacy protection for researchers. Moreover, current systems lack an integrated mechanism for decentralized reputation tracking and privacy-preserving identity verification.

To address these limitations, this paper proposes DeBug, a hybrid decentralized bug bounty platform designed to enhance transparency, fairness, and privacy in vulnerability disclosure programs. The proposed system integrates smart contracts, decentralized storage, and cryptographic identity verification to create a secure and efficient ecosystem for ethical hacking collaboration.

II. LITERATURE REVIEW

Reference [1] proposes Bountychain, a blockchain-based decentralized bug bounty platform that leverages Ethereum smart contracts and InterPlanetary File System (IPFS) for secure vulnerability reporting. The system automates reward payouts and ensures transparency by recording vulnerability submissions and payments on the blockchain. However, the approach faces challenges related to Ethereum gas fees and scalability issues when handling large vulnerability reports.

Reference [2] introduces VeriOSS, a blockchain-driven vulnerability disclosure framework designed to foster fair collaboration between organizations and security researchers. However, the framework assumes a limited participant model and lacks strong identity verification mechanisms for researchers.

Reference [3] presents an empirical analysis of bug bounty programs across major platforms such as HackerOne and Bugcrowd. The study evaluates the effectiveness of crowdsourced vulnerability discovery and highlights how bug bounty programs improve software security by enabling large-scale participation from independent researchers. However, the study identifies limitations including delayed payouts, limited transparency in vulnerability triage, and dependency on intermediaries.

Reference [4] explores the application of blockchain technology in collaborative cybersecurity systems. The proposed framework demonstrates how distributed ledger technologies can improve trust, transparency, and accountability in vulnerability reporting ecosystems. While the system ensures secure data sharing between stakeholders, it does not address privacy-preserving identity verification for security researchers.

Reference [5] investigates decentralized storage systems such as IPFS for secure data management in distributed applications. The study highlights how content-addressable storage and cryptographic hashing can ensure data integrity, immutability, and resistance to tampering. These properties make decentralized storage solutions suitable for handling sensitive data such as vulnerability reports.

Reference [6] analyzes the role of smart contract optimization in improving blockchain application performance. The research introduces techniques such as storage minimization, event-based logging, and efficient data structures to reduce gas consumption and improve execution efficiency in blockchain-based systems.

The proposed work builds upon these research contributions by integrating blockchain-based escrow mechanisms, decentralized storage, and privacy-preserving identity verification into a unified bug bounty platform. By combining smart contracts, IPFS-based report storage, and decentralized reputation tracking, the system aims to improve transparency, scalability, and trust in vulnerability disclosure ecosystems.

III. NOVELTY AND IMPACT

The proposed system DeBug introduces a decentralized bug bounty ecosystem designed to address several limitations present in traditional vulnerability disclosure platforms. Existing bug bounty systems often rely on centralized infrastructure to manage vulnerability submissions, validate reports, and distribute rewards. While such platforms have successfully enabled crowdsourced security testing, they introduce several challenges, including delayed payouts, lack of transparency in triage processes, and privacy concerns for researchers. DeBug aims to overcome these challenges by integrating blockchain-based smart contracts, decentralized storage systems, and privacy-preserving identity mechanisms. By leveraging decentralized technologies, the platform eliminates the need for intermediaries and creates a transparent environment for vulnerability reporting and reward distribution.

The impact of DeBug can be categorized into direct impacts, which immediately improve the vulnerability disclosure process, and indirect impacts, which influence the broader cybersecurity ecosystem.

Impact Analysis:

The introduction of DeBug has the potential to significantly influence the cybersecurity ecosystem by improving vulnerability disclosure processes and researcher collaboration. The impacts can be categorized into direct impacts and indirect impacts.

1. Direct Impacts -

1.1 Transparent and Automated Reward Distribution:

One of the major challenges in traditional bug bounty platforms is the delay in reward payouts. On centralized platforms, vulnerability submissions typically go through several stages including triage, validation, and administrative approval before rewards are processed. These manual processes can cause significant delays and may also lead to disputes between organizations and researchers regarding payout decisions.

DeBug addresses this issue by implementing smart contract-based escrow mechanisms. When an organization creates a bounty program, the reward amount is locked within a blockchain smart contract. This ensures that the funds are secured and available for payout before vulnerability submissions are accepted.

Once a vulnerability report is validated, the smart contract automatically releases the reward to the researcher's cryptocurrency wallet. This automation eliminates manual payment processing, reduces disputes, and ensures transparent and fair compensation for ethical hackers.

1.2 Secure Storage of Vulnerability Reports:

Vulnerability reports often contain sensitive technical details about security weaknesses in systems or applications. Storing such information on centralized servers can introduce risks such as data breaches, unauthorized access, or accidental data loss.

To address this challenge, DeBug utilizes InterPlanetary File System (IPFS) for decentralized storage of vulnerability reports. IPFS stores files using content-addressable hashing, which means each report is assigned a unique cryptographic hash that verifies its integrity.

By storing encrypted vulnerability reports on IPFS and maintaining file references on the blockchain, DeBug ensures that vulnerability data remains tamper-proof, secure, and permanently accessible. This decentralized approach eliminates single points of failure and significantly improves the security of vulnerability disclosure systems.

1.3 Privacy-Preserving Researcher Participation:

Many bug bounty platforms require security researchers to complete extensive identity verification procedures before receiving rewards. While such processes help prevent fraudulent activities, they can discourage participation from independent researchers who prefer to maintain anonymity.

DeBug integrates Zero-Knowledge (ZK) identity verification mechanisms to address this issue. Zero-Knowledge proofs allow users to prove certain attributes about themselves without revealing sensitive personal information.

Through this approach, researchers can participate in vulnerability disclosure programs pseudonymously while still maintaining credibility within the platform. This privacy-preserving model encourages broader participation from ethical hackers while ensuring trust and accountability within the ecosystem.

2. Indirect Impacts -

2.1 Encouraging Ethical Hacking Participation:

Bug bounty programs rely heavily on the participation of independent security researchers. However, many ethical hackers hesitate to participate in vulnerability disclosure programs due to concerns regarding unfair payouts, lack of transparency in vulnerability validation, and restrictive identity verification processes. These barriers can reduce the number of researchers actively contributing to vulnerability discovery.

By introducing transparent smart contract-based reward mechanisms and privacy-preserving participation, DeBug creates an environment that encourages greater involvement from the global security community. Researchers can submit vulnerabilities with confidence knowing that reward distribution is automated and governed by blockchain rules rather than centralized intermediaries.

As a result, DeBug has the potential to significantly increase the participation of ethical hackers in vulnerability disclosure programs. Greater participation improves the chances of identifying security vulnerabilities earlier, ultimately strengthening the overall security posture of digital systems.

2.2 Strengthening Trust Between Organizations and Researchers:

Trust plays a critical role in the success of bug bounty ecosystems. In many traditional platforms, the vulnerability triage process is controlled by centralized administrators, which can create uncertainty for researchers regarding how their submissions are evaluated or why certain reports are rejected. DeBug improves trust by recording key events in the vulnerability disclosure process on the blockchain. Actions such as bounty creation, vulnerability submission timestamps, and reward payouts are stored as immutable records that can be verified by all stakeholders.

This transparency ensures that both organizations and researchers can rely on a verifiable and auditable system. By reducing ambiguity in the vulnerability validation process, DeBug strengthens collaboration between organizations seeking security improvements and researchers contributing their expertise.

2.3 Advancing Decentralized Cybersecurity Infrastructure:

The integration of blockchain technology into cybersecurity platforms represents a growing area of innovation. DeBug contributes to this trend by demonstrating how decentralized technologies can be applied to vulnerability disclosure systems.

By combining smart contracts, decentralized storage, and cryptographic identity verification, the platform introduces a new architectural model for bug bounty programs. This hybrid approach balances the transparency of blockchain with the scalability of off-chain systems.

Such developments may inspire further research and innovation in decentralized cybersecurity frameworks. In the future, similar approaches could be applied to other areas such as incident response platforms, threat intelligence sharing systems, and decentralized security auditing tools.

2.4 Promoting Global Accessibility in Vulnerability Disclosure:

Traditional bug bounty platforms often face limitations when handling international payments to researchers. Differences in banking systems, currency conversions, and transaction fees can create delays and complications when distributing rewards globally. DeBug addresses these issues by utilizing blockchain-based cryptocurrency payments. Since blockchain transactions operate independently of traditional financial institutions, researchers from different regions can receive rewards quickly and securely.

This global accessibility enables researchers from developing regions or underrepresented communities to participate in vulnerability disclosure programs. As a result, organizations can benefit from a more diverse and globally distributed pool of security expertise.

2.5 Encouraging Innovation in Security Platforms:

The introduction of decentralized technologies in vulnerability management platforms can influence the development of future cybersecurity tools. By demonstrating how blockchain can improve transparency, security, and automation, DeBug provides a foundation for building more advanced decentralized security applications. Developers and researchers may build upon this framework to create new systems that integrate automated vulnerability analysis, decentralized governance models, or AI-assisted vulnerability triage mechanisms.

Such innovations could lead to the development of next-generation cybersecurity ecosystems where organizations, researchers, and automated security tools collaborate in a transparent and decentralized environment.

IV. PROPOSED SOLUTION

A. Overview

The proposed system DeBug is a decentralized bug bounty platform designed to improve transparency, security, and efficiency in vulnerability disclosure programs.

The overall architecture of the system is illustrated in Fig. 4a.1 (Block Diagram). The architecture consists of four major components: the user interface layer, client layer, API gateway, and blockchain-based payment layer.

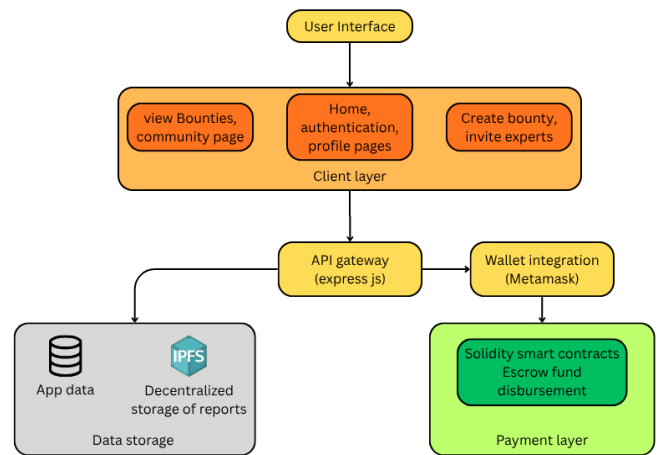


fig 4a.1

The user interface layer provides the primary interaction point for users. Through this interface, researchers can explore bounty programs, submit vulnerability reports, and track their submissions, while organizations can create bounty programs and manage vulnerability reports.

The client layer manages core platform functionalities such as authentication, profile management, and bounty management pages. This layer interacts with the backend services through secure API calls.

The API gateway, implemented using Express.js, acts as the bridge between the frontend application and backend services. It handles authentication, request validation, and communication with decentralized storage systems and blockchain services.

The blockchain payment layer uses smart contracts to manage escrow funds and automate bounty payouts. Vulnerability reports are stored in IPFS, ensuring decentralized and tamper-proof storage, while application data is stored in the backend database for efficient retrieval.

B. Modular Architecture

The internal system modules and their interactions are illustrated in Fig. 4b.1 (Modular Diagram). The platform is divided into several functional modules to ensure scalability.

The User Module handles user roles and authentication. The platform supports multiple roles including researchers, project administrators, and organizations. Users can create profiles and authenticate through wallet-based login mechanisms. The Project Module manages bounty project registration and scope definition. Organizations can define vulnerability scopes, reward tiers, and submission deadlines for each bounty program.

The Bounty Management Module allows organizations to create, update, and close bounty programs. During bounty creation, funds are locked in smart contracts to ensure reward availability. The Submission Module manages encrypted vulnerability report submissions. Researchers submit vulnerability reports through secure channels, and the system records the submissions for validation and review.

The Payment Module handles reward distribution using smart contracts. Once a vulnerability report is approved, the smart contract automatically releases the bounty reward to the researcher. The Analytics Module provides dashboards for monitoring submissions, tracking payments, and analyzing platform activity. Additionally, the Governance Module enables community-based dispute resolution mechanisms through decentralized voting.

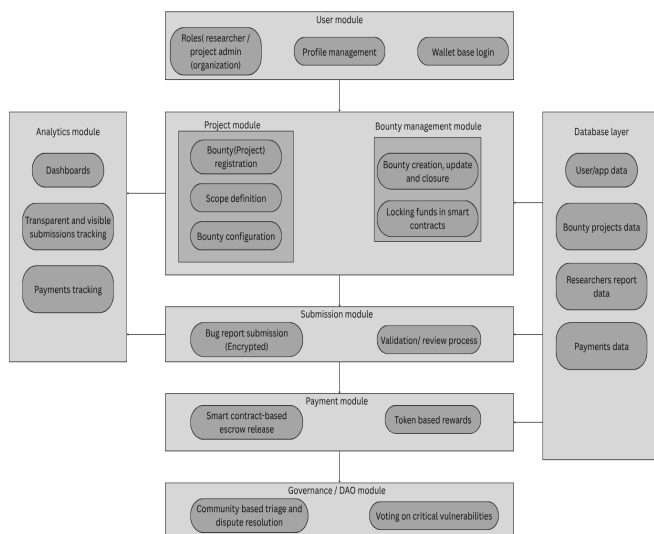


fig 4b.1

C. Data Flow Design

The flow of information in the DeBug platform is represented through Data Flow Diagrams (DFD).

C.1 DFD Level-0

The Level-0 DFD, shown in Fig. 4c.1, illustrates the high-level interaction between system entities and the platform.

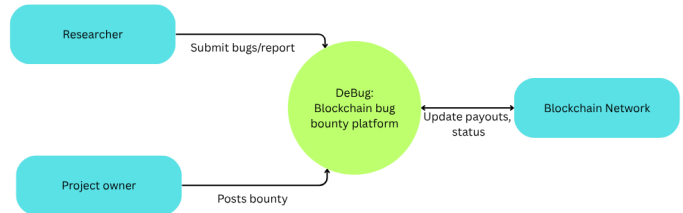


fig 4c.1

Two primary entities interact with the system: security researchers and project owners (organizations). Researchers submit vulnerability reports to the DeBug platform, while organizations create bounty programs and review submitted reports. The platform processes these actions and communicates with the blockchain network to update payment status and transaction records.

This high-level data flow shows how the DeBug platform acts as the central hub connecting researchers, organizations, and blockchain services.

C.2 DFD Level-1

The Level-1 DFD, shown in Fig. 4c.2, provides a more detailed representation of the internal processes within the platform.

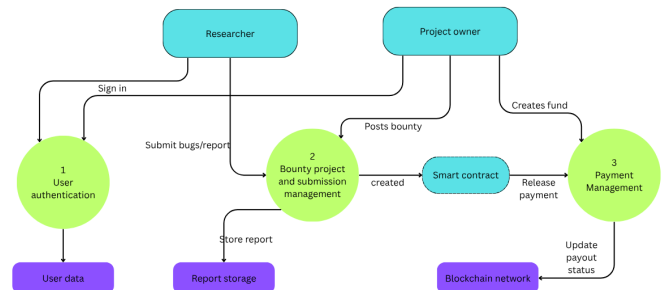


fig 4c.2

The system consists of three primary processes:

User Authentication – Users authenticate using secure login mechanisms. User data is stored in the platform database.

Bounty Project and Submission Management – Organizations create bounty programs, and researchers submit vulnerability reports. These reports are securely stored and tracked by the platform.

Payment Management – Once a vulnerability is validated, the smart contract releases the reward to the researcher's wallet and updates the transaction status on the blockchain network.

This layered data flow ensures that the system maintains a clear separation between authentication, vulnerability management, and payment processes.

D. Workflow of the Vulnerability Disclosure Process

The operational workflow of the platform is illustrated in Fig. 4d.1 (Flowchart).

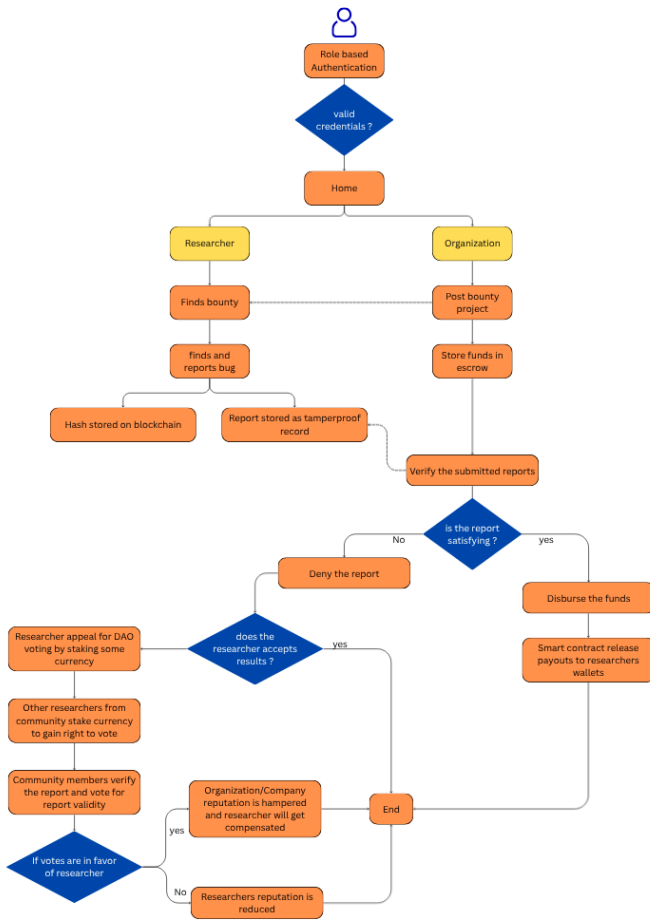


fig 4d.1

The process begins with role-based authentication, where users log in as either researchers or organizations. Organizations create bounty programs by defining vulnerability scopes and depositing funds into a smart contract escrow. These funds remain locked until a valid vulnerability is discovered.

Researchers browse available bounty programs and submit vulnerability reports when a potential security flaw is identified. The report is encrypted and stored as a tamper-proof record in the decentralized storage system.

The organization reviews the submitted report and verifies whether the vulnerability meets the bounty criteria. If the report is valid, the smart contract automatically disburses the reward to the researcher's wallet.

In cases where the report is rejected and the researcher disagrees with the decision, the system provides a community-based dispute resolution mechanism through decentralized voting. Community members review the report and vote on its validity.

This workflow ensures transparency, fairness, and accountability in the vulnerability disclosure lifecycle.

E. Key System Components

The DeBug platform integrates several key technologies to ensure efficient system operation. Blockchain Smart Contracts are used to manage bounty escrow accounts and automate reward payouts. Decentralized Storage (IPFS) ensures secure and tamper-resistant storage of vulnerability reports. Backend Services manage authentication, report validation, and platform analytics. Frontend Applications provide intuitive interfaces for both researchers and organizations.

Together, these components form a scalable and transparent ecosystem for decentralized vulnerability disclosure.

V. OPTIMISATION AND STRATEGY

The DeBug platform is designed with several optimization strategies to ensure scalability, efficiency, and security across its decentralized infrastructure.

A. Smart Contract Optimization

Smart contract execution on blockchain networks can incur computational costs and latency. To minimize these issues, DeBug implements optimized smart contract logic that reduces unnecessary storage operations and relies on event-based logging for tracking transactions.

By storing only essential metadata on the blockchain and offloading large data storage to decentralized systems such as IPFS, the platform significantly reduces on-chain storage requirements. This approach improves transaction efficiency while maintaining transparency.

B. Hybrid On-Chain and Off-Chain Architecture

Storing all vulnerability data directly on the blockchain would lead to scalability issues and increased transaction costs. To address this challenge, DeBug adopts a hybrid architecture that separates critical trust mechanisms from large data storage.

Blockchain components handle escrow management, payout transactions, and reputation tracking, while off-chain infrastructure manages authentication, analytics, and report metadata storage. This hybrid design allows the system to maintain transparency and security while ensuring scalability for large-scale vulnerability disclosure programs.

C. Performance and User Experience Improvements

User experience plays an important role in encouraging participation from both organizations and security researchers. The DeBug platform provides an intuitive interface that simplifies bounty creation, vulnerability submission, and submission tracking.

Asynchronous processing mechanisms allow vulnerability submissions and blockchain transactions to be handled in the background while users continue interacting with the platform. This reduces perceived latency and improves overall usability.

In addition, caching techniques and optimized database queries are used to improve platform responsiveness and ensure efficient retrieval of bounty information and researcher statistics.

ACKNOWLEDGMENT

We would like to express our deepest gratitude to Ms. Geocey Shejy, our project guide, for her constant support, valuable guidance, and insightful feedback throughout the development of this research. Her expertise and encouragement have played a significant role in shaping our approach and refining the ideas presented in this study.

We also extend our sincere appreciation to Dr. Nupur Giri, Head of the Computer Engineering Department, for providing us with the opportunity and resources to carry out this research work. Her support and academic leadership have been instrumental in motivating us to pursue this project and explore innovative solutions in the field of cybersecurity and blockchain technology.

Additionally, we would like to acknowledge the contributions of the blockchain and cybersecurity communities, whose pioneering research and open-source technologies have inspired the development of decentralized security platforms such as DeBug. The availability of decentralized technologies such as blockchain frameworks, smart contract platforms, and distributed storage systems has played a crucial role in the implementation and analysis of our proposed system.

Finally, we extend our sincere thanks to our peers, faculty members, and family members for their continuous encouragement and support throughout the course of this project. Their motivation and constructive feedback have greatly contributed to the successful completion of this research work.

CONCLUSION AND FUTURE WORK

This project represents a significant step toward improving the transparency, security, and efficiency of bug bounty ecosystems by leveraging decentralized technologies. By integrating blockchain-based smart contracts, decentralized storage systems, and privacy-preserving identity mechanisms, the proposed DeBug platform establishes a trustless environment for vulnerability disclosure and reward distribution. Through the use of smart contract-based escrow mechanisms, the platform ensures automated and transparent bounty payouts once vulnerabilities are validated. This eliminates delays commonly associated with manual reward processing and reduces disputes between organizations and security researchers. Additionally, the implementation of decentralized storage using IPFS ensures that vulnerability reports remain tamper-proof, secure, and permanently accessible.

Another key improvement introduced by this system is the integration of a decentralized reputation mechanism, which allows security researchers to build verifiable credibility through validated vulnerability submissions. This encourages ethical participation and enables organizations to identify experienced researchers for future security programs. The hybrid architecture combining blockchain-based trust mechanisms and scalable off-chain infrastructure enables the platform to maintain transparency while supporting large-scale vulnerability disclosure programs. By balancing decentralization with efficient system design, the proposed solution demonstrates how blockchain technology can enhance collaborative cybersecurity practices.

Beyond technical implementation, the platform also emphasizes usability and accessibility. The intuitive interface for bounty creation, vulnerability submission, and report tracking allows both organizations and researchers to interact with the system seamlessly. These improvements enhance user engagement and promote broader adoption of decentralized vulnerability disclosure platforms.

Future Work:

To further enhance the capabilities of the DeBug platform, several future improvements can be explored –

- Integration of DAO-based governance mechanisms, enabling decentralized voting for vulnerability triage and dispute resolution, allowing the community to participate in validating critical vulnerability reports.
- Multi-chain interoperability, allowing bounty programs and reward payouts to operate across multiple blockchain networks such as Ethereum, Polygon, and other compatible ecosystems.
- AI-driven vulnerability triage systems, which can assist organizations in automatically prioritizing vulnerability reports based on severity, exploitability, and potential impact.
- Enhanced security measures, including multi-signature smart contract authorization and periodic security audits to further strengthen the reliability and integrity of the platform.
- Advanced analytics and reporting dashboards, providing organizations with deeper insights into vulnerability trends, researcher activity, and security performance metrics.

In conclusion, the DeBug platform demonstrates how decentralized technologies can transform vulnerability disclosure ecosystems by improving transparency, reducing operational friction, and encouraging ethical hacker participation. By enhancing trust, security, and collaboration, the system contributes to the development of a more scalable and efficient future for decentralized cybersecurity platforms.

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